Zonestar

Slicing Guide
for mixing Color print

(Base on Cura 4.7 or later)

V1.0

Contents

- Download and install Cura
- Import Zonestar Printer Settings
- Setting up printer
- Setting up filament
- Slicing singel color 3d object
- Slicing 2 color 3d object
- Slicing mullti colors 3d object(used colors <= actual extruders of printer)
- Slicing more colors 3d object by using virual extruder(used colors > actual extruders of printer, now it is up to 8 colors)

Download and install Cura

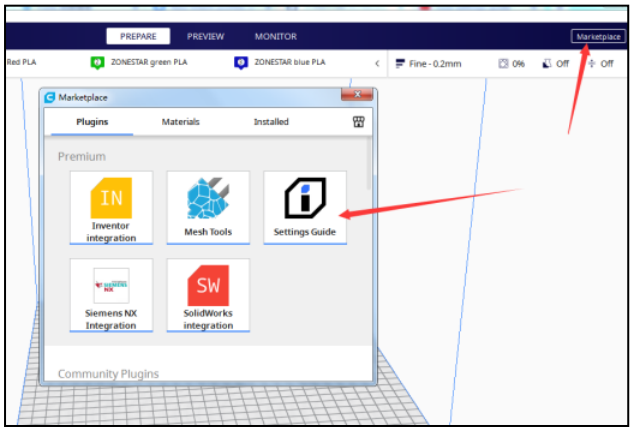
Download cura from the below link and install it to your PC:

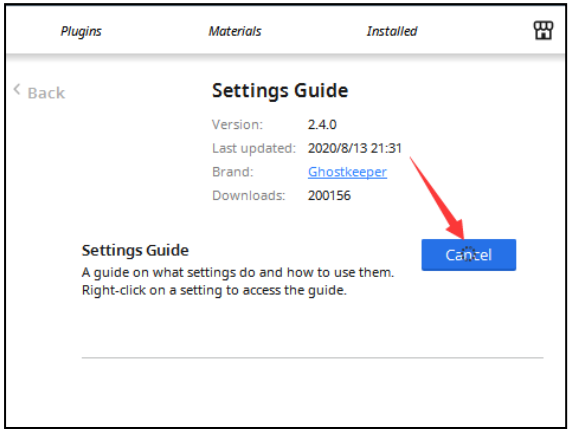
https://ultimaker.com/software/ultimaker-cura

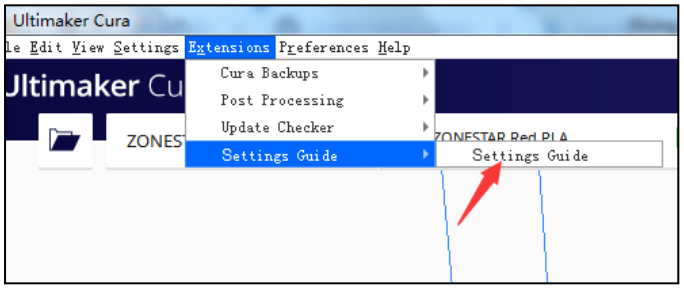
About how to install and use Cura, please refer to this link:

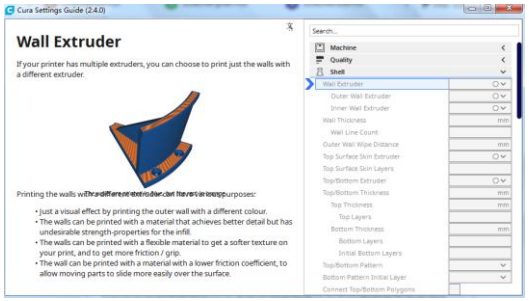
https://support.ultimaker.com/hc/en-us/categories/360002327600

If you want to know more about the settings of cura, please install a “settings guide” plugin in cura, and then open it to study:



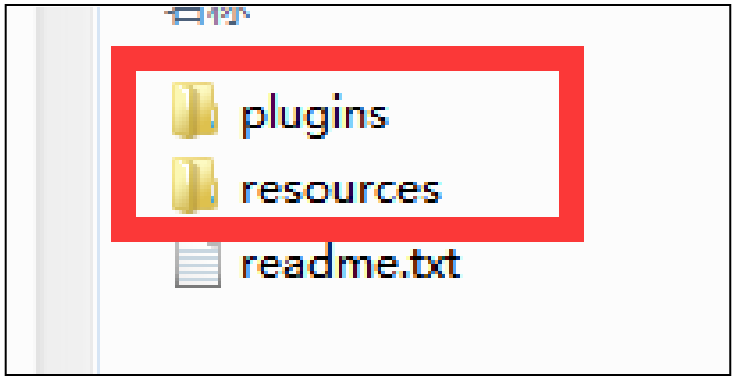


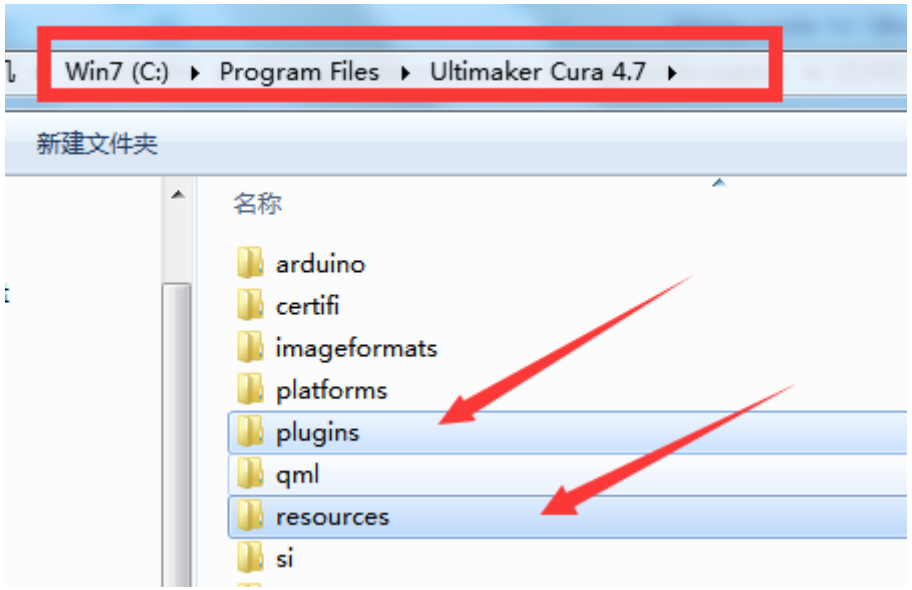




Import ZONESTAR printer settings

1. Download "zonestar Cura Resources “ from the below link:
   Download link: https://github.com/ZONESTAR3D/Slicing-Guide
2. Copy the contents of this sirctory to the installation directory of Cura

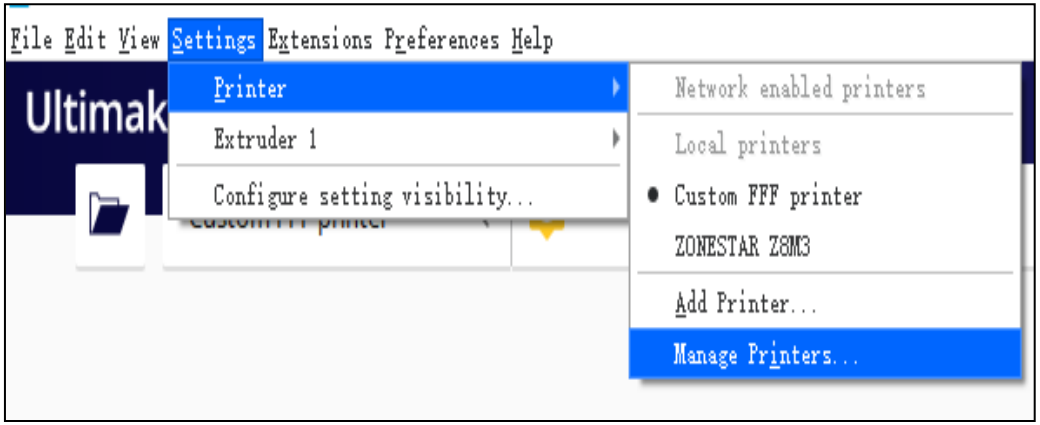




3. Run the cura

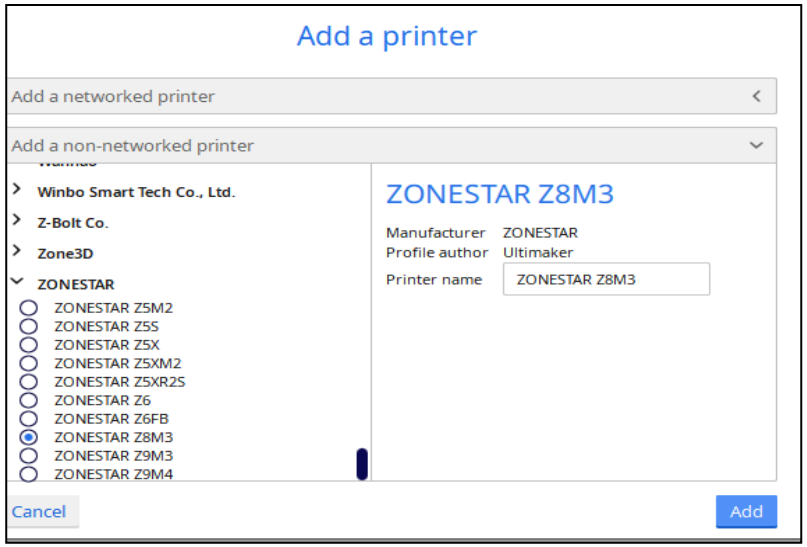
Setting up printer

1: Open “Manage printers”

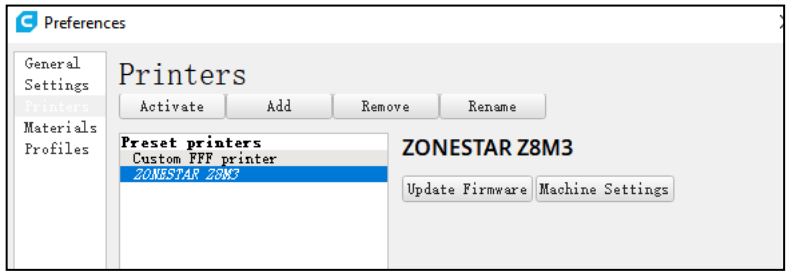


2: Click “Add”

3: Select your printer model,then click “Add”



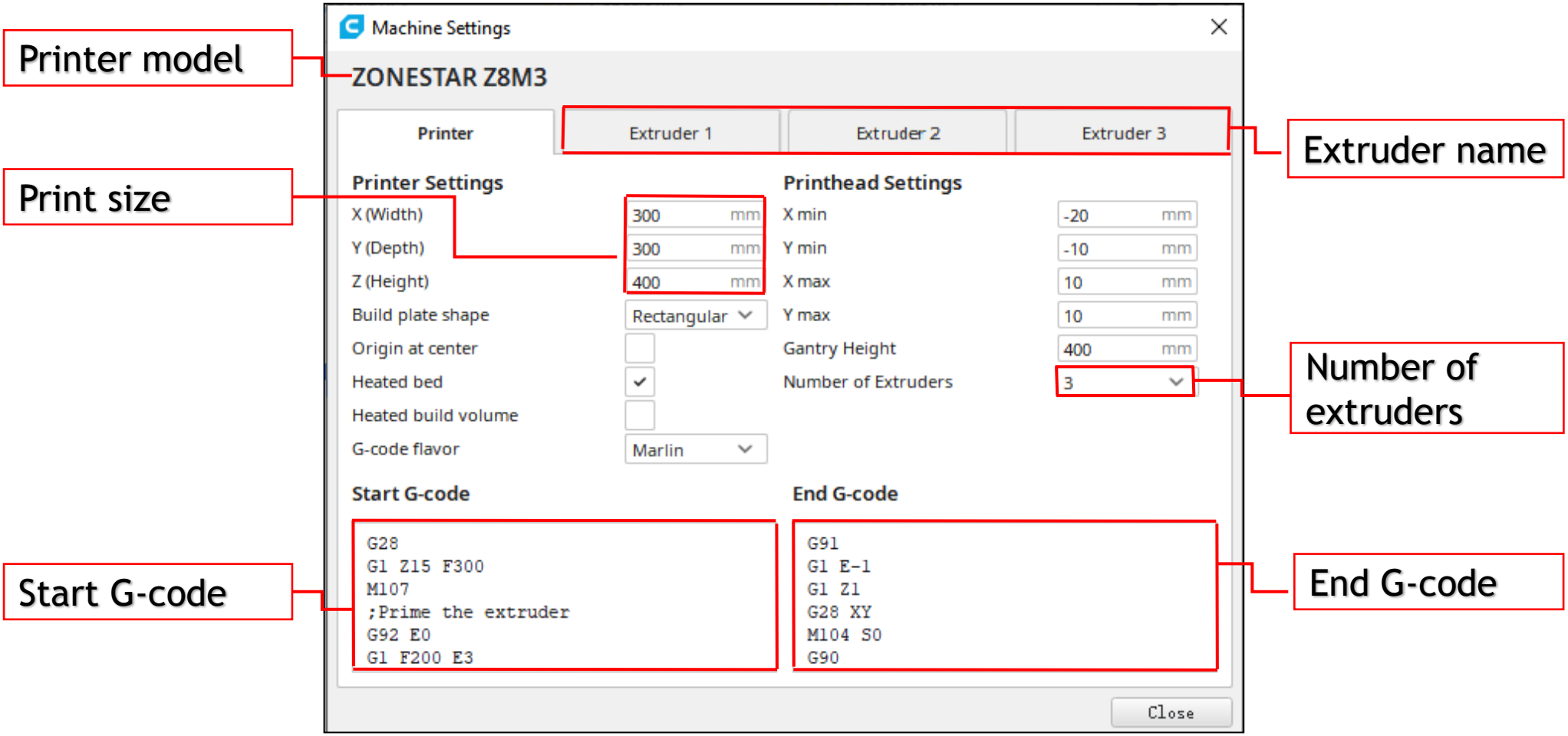
4: Click “Activate”



zonestar

Setting up filament

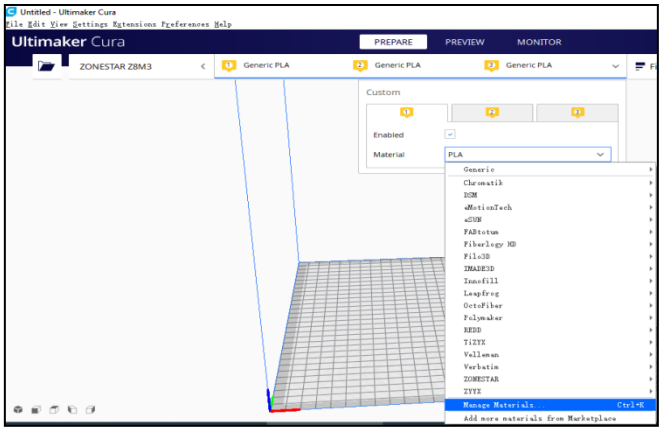
Click “Machine settings”. Check the printer parameters.



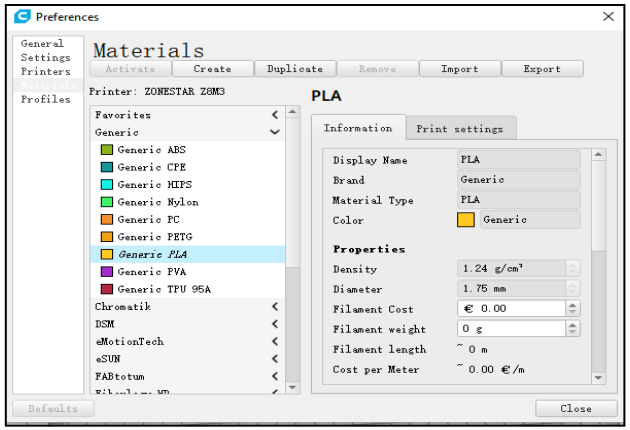
Setting up filament

In order to easy to view when slicing, you can define the filament color

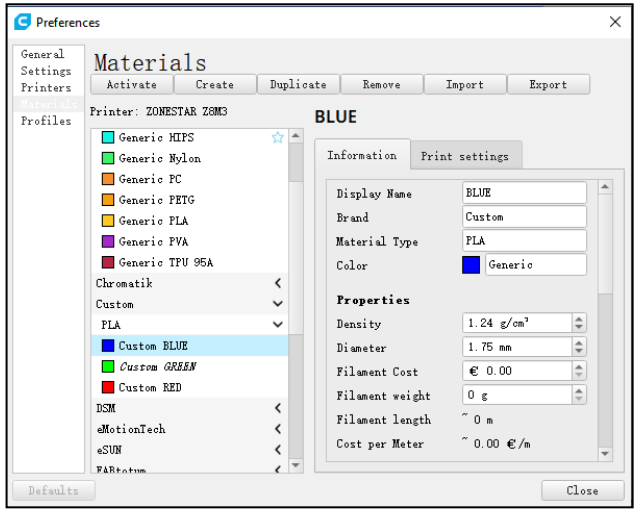
1: Open “Manage materials...”



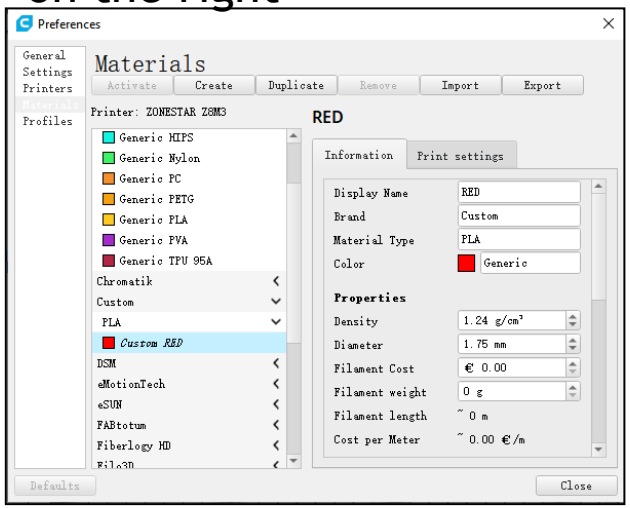
2: Click “Create”



4:Click “Duplicate”,and define three colors

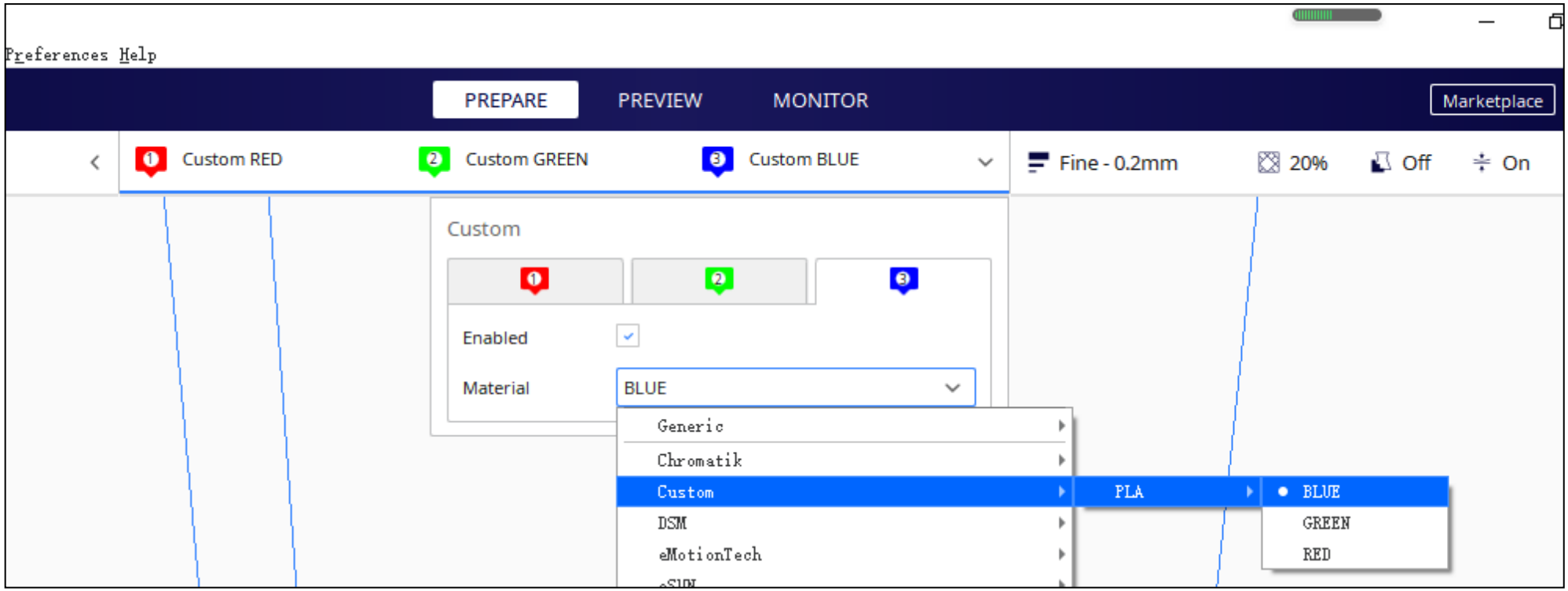


3:Find custom and modify relevant parameters in the information column on the right



Setting up filament

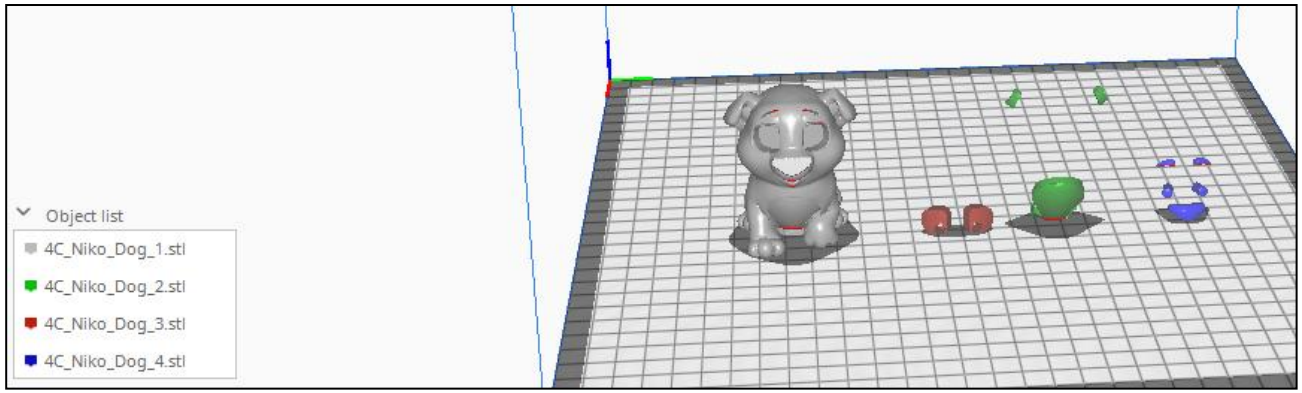
Select the filament color in turn on the main page.As shown in the figure below.



You can choose ZONESTAR filament too, we have set some color for PLA and PETG

Slicing

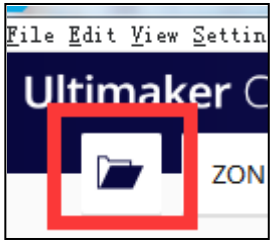
For the sake of illustration, we will use only one 3d object in the following pages. This 3d object is a 4-color model, which has divided the object into four parts

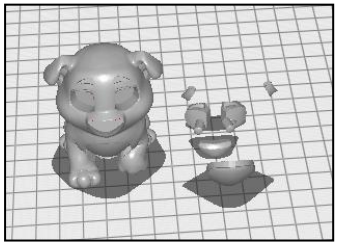


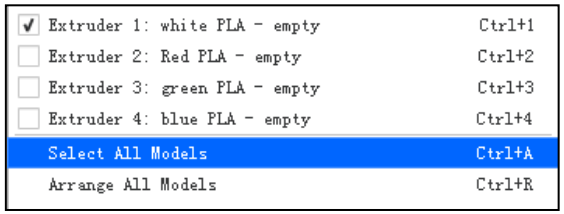
If you need to print multicolor, you need a 3d object that has been divided (the number of divided parts is according to the number of colors), and their origin position must be consistent in order to be merged.

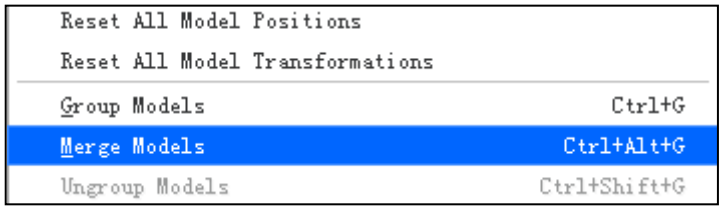
Of course, you can also merge several objects into one color (multiple parts are assigned to the same extruder), as you will see in the next pages

Slicing for singel color 3d object printing





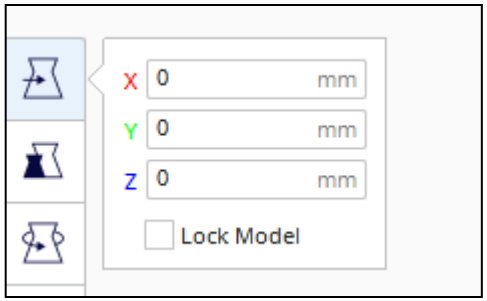


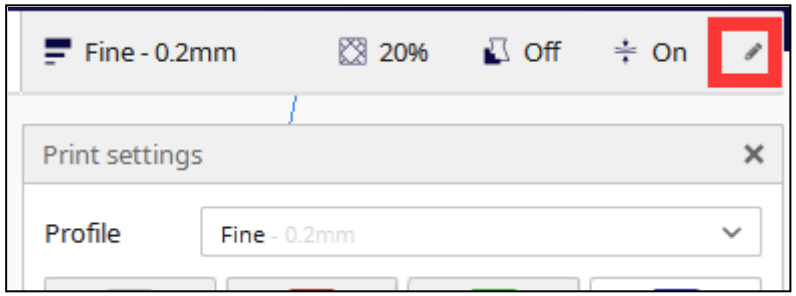


Load files

Right click the mouse
Setlect all models

Merge

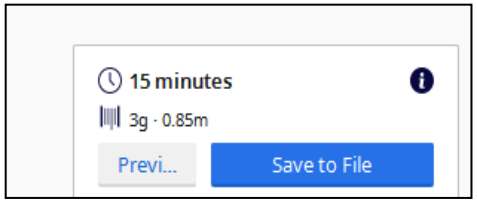




Move/Scale/Rotat
e the model

Set slicing parameter
NOTE: load “1C_Niko_dog.3mf” to see reference settings

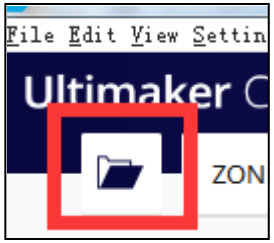
Slicing it

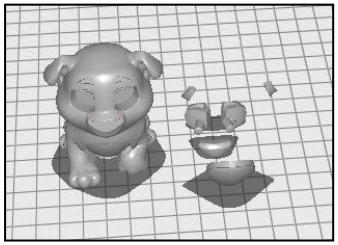


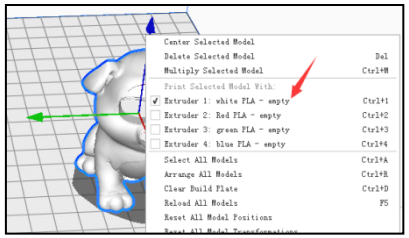
save it

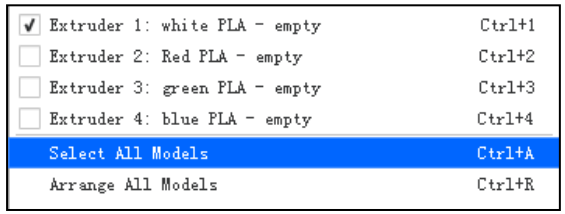
Copy the gcode file to SD card, then print it

Slicing 2 Color 3d object





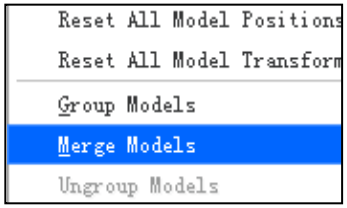


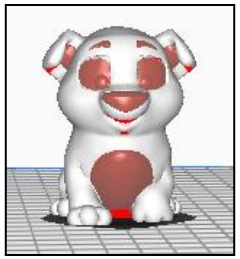


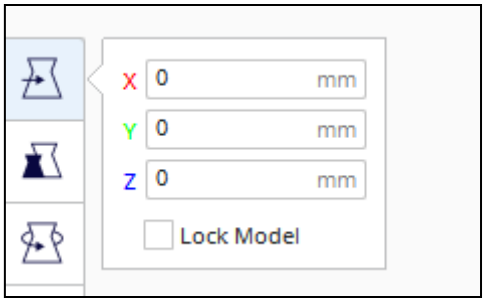
Load files

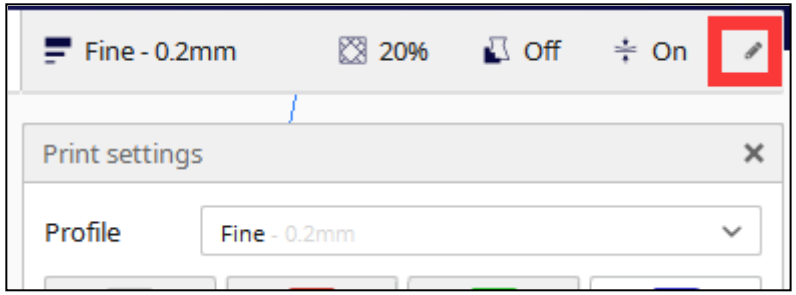
Right click the part and assign extruder for it

Right click the mouse Setlect all models





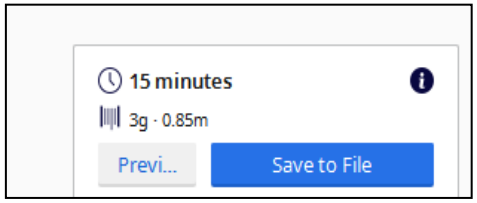




Merge

Move/Scale/Rotat e the model

Set slicing parameter



Slicing it

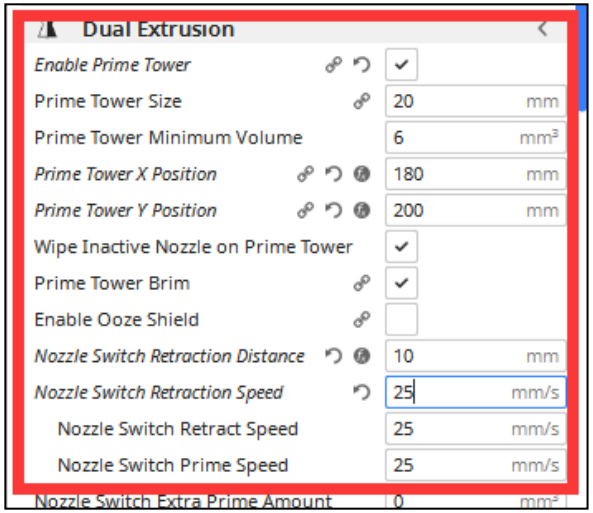
save it

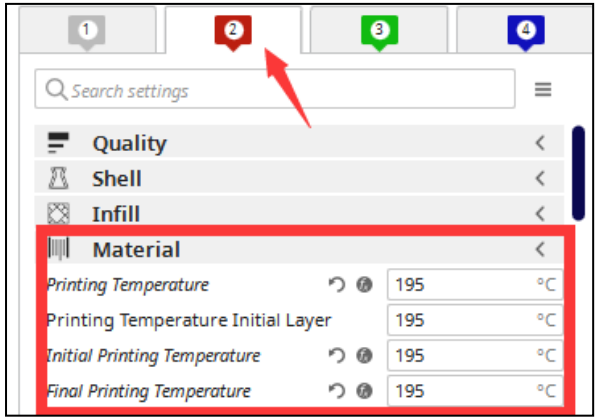
Copy the gcode file to SD card, then print it

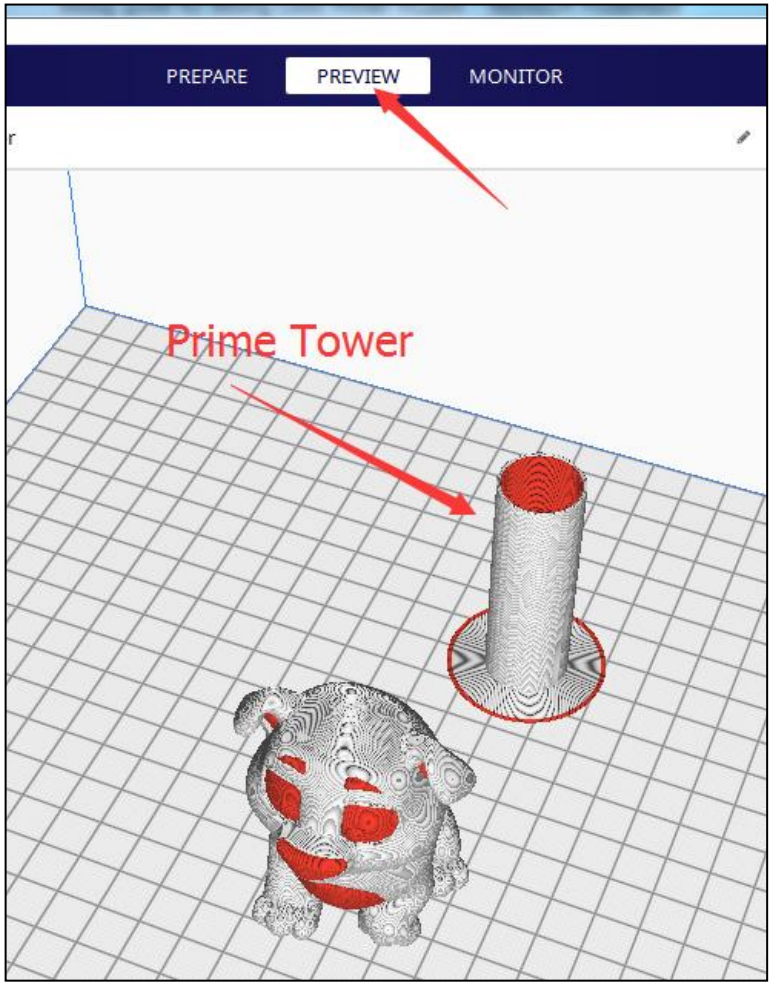
Slicing setting for 2 color printing

We need to set the below setting for 2 color printing:

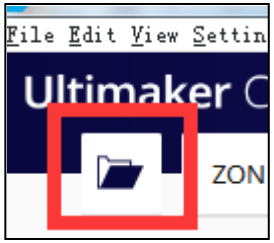
1. Enable a prime tower and set its position
2. Endable nozzle swicht retraction parameters (need to set in both of extruders)
3. Set the filament temperature for the 2nd extruder (set to the same with 1st extruder)

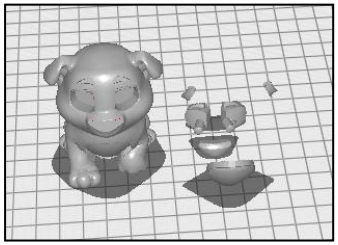


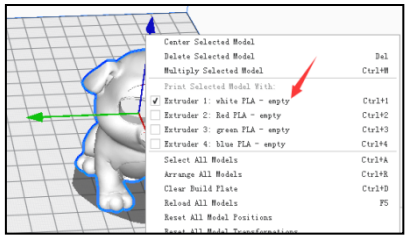


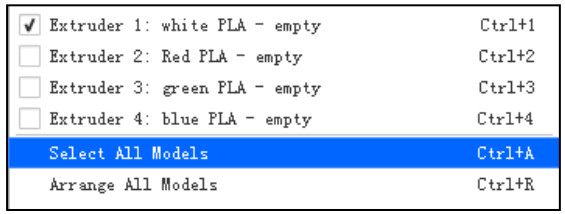


Slicing multi colors 3d object





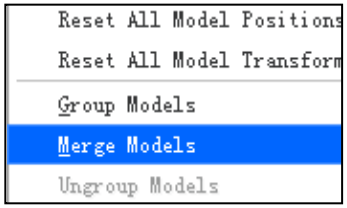


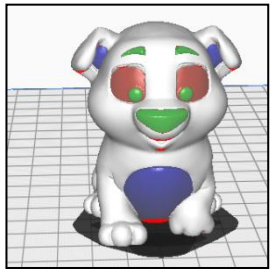


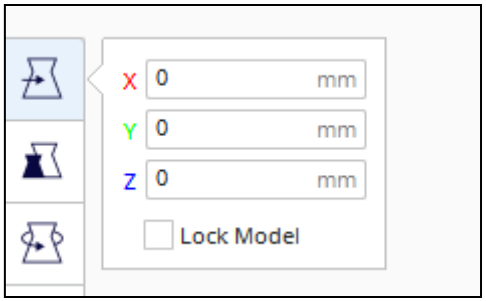
Load files

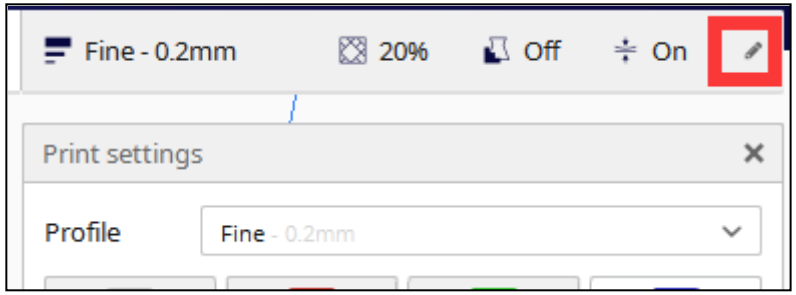
Right click the part and assign extruder for each

Right click the mouse Setlect all models





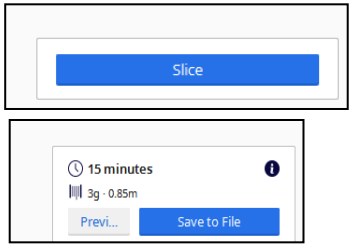


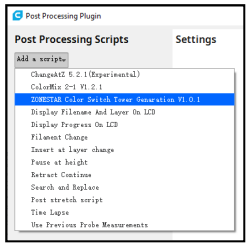


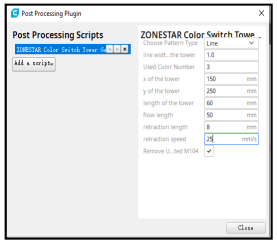
Merge

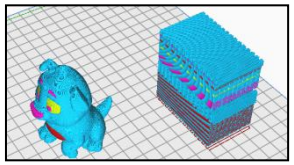
Move/Scale/Rotat e the model

Set slicing parameter









slicing and save it to PC

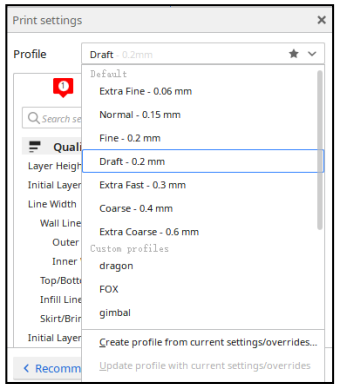
Add Color Swith Tower

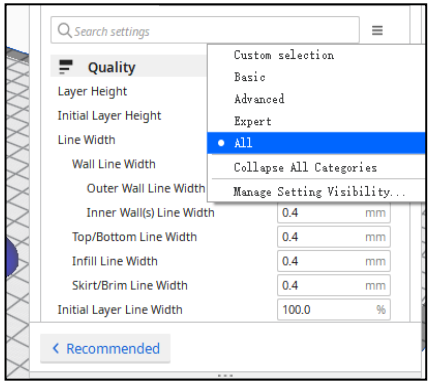
Copy to SD card and print it

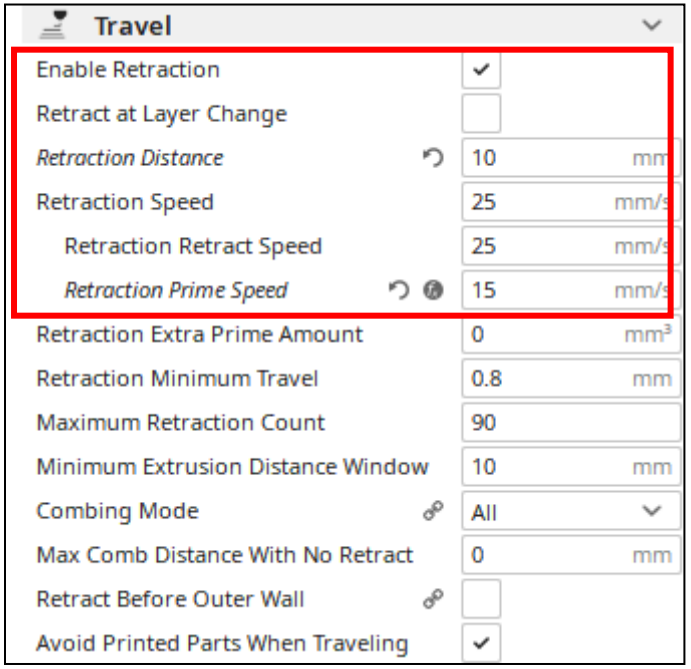
Slicing multi colors 3d object - slicing

NOTE: When printing settings, please note that it needs to be set for each extruder.

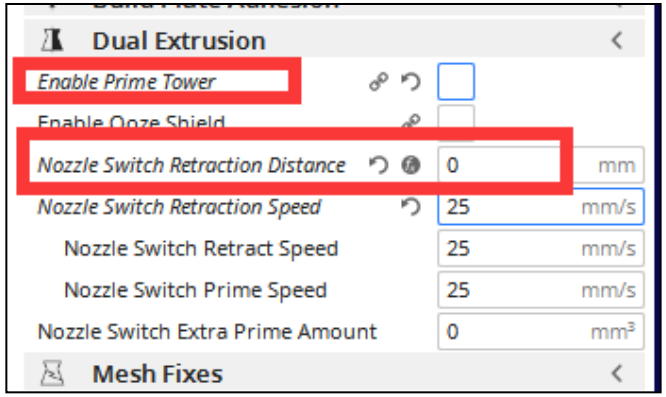
NOTE: The below settings are for PLA filament, if you want to choose other type of filament, please modify the nozzle temperature hotbed temperature to correct value

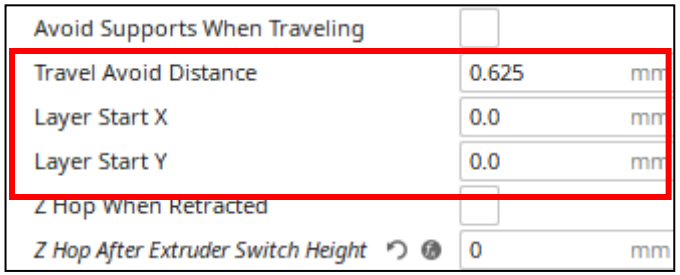






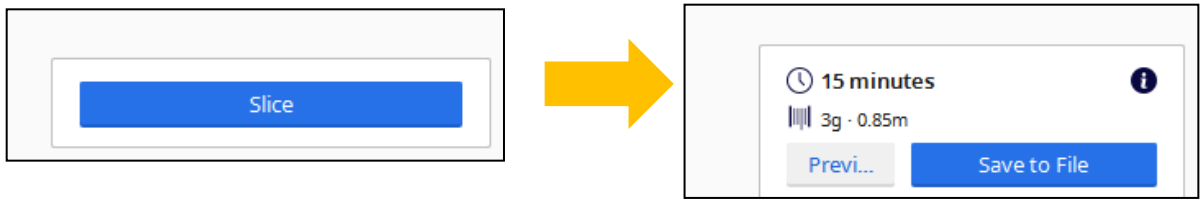
NOTE: Pay attention to the retraction parameters in "travel" and "dual Extrusion", as below:



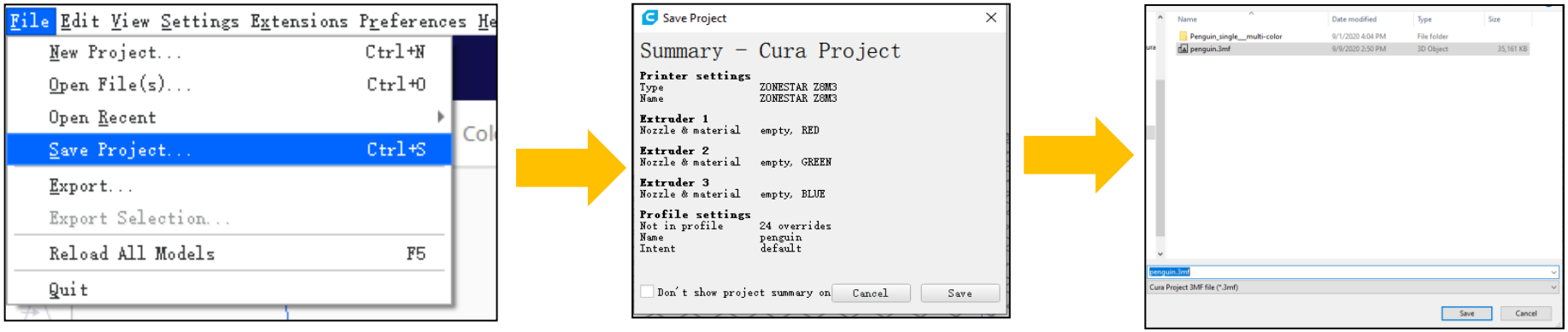


Slicing multi colors 3d object - Slice and Save

When all settings are completed, slicing and store the gcode file to your PC.



We suggest you save the project file for the next call. All configurations will follow the previous settings.

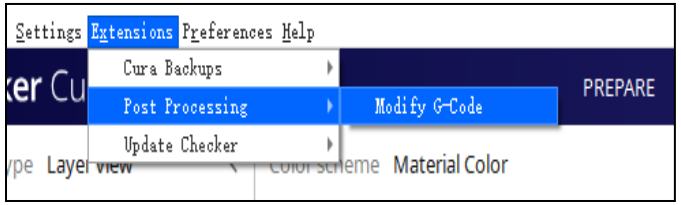


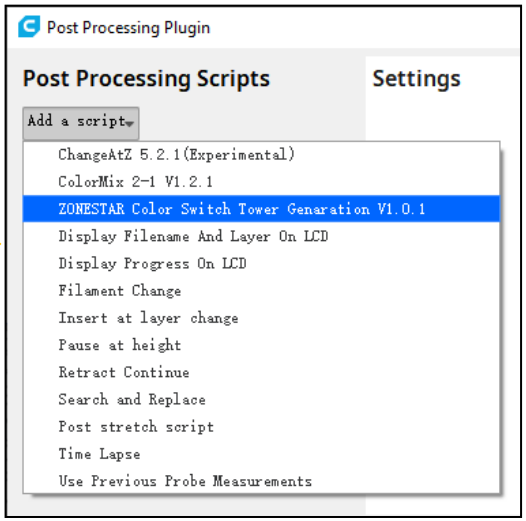
!NOT FINISHED!

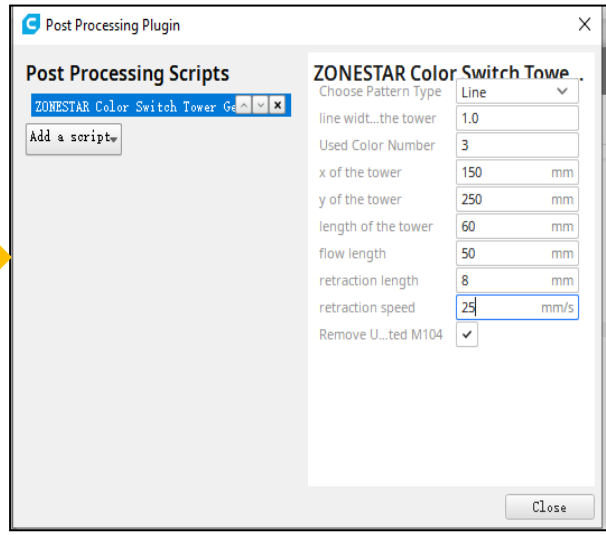
Cura can't support more than 2 color by mixing color extruder, we need to *use ZONESTAR ColorSwitchTower plugin* to process the sliced gcode files.

Slicing multi colors 3d object - Genarate color switching tower

1.Open plug-in.



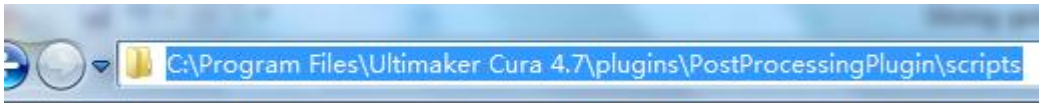


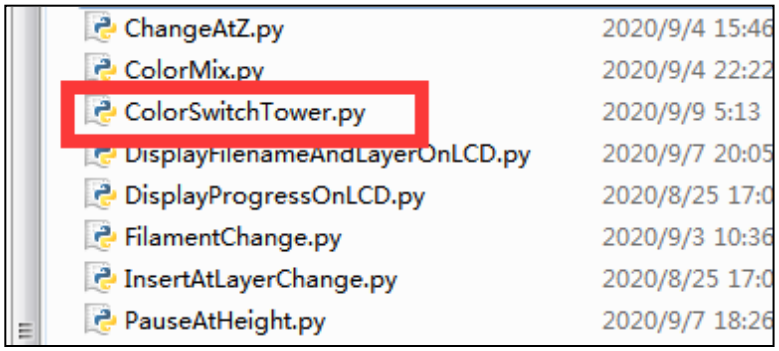


NOTE: If you can't find the "ZONESTAR Color Switch Tower Generation" plug-in in the list, please check if there is a "ColorSwitchTower.py" in the below directory of your PC:

%install directory of Cura%\plugins\PostProcessingPlugin\scripts

If there isn't, please refer to "page 4" to download it and copy to the above directory.

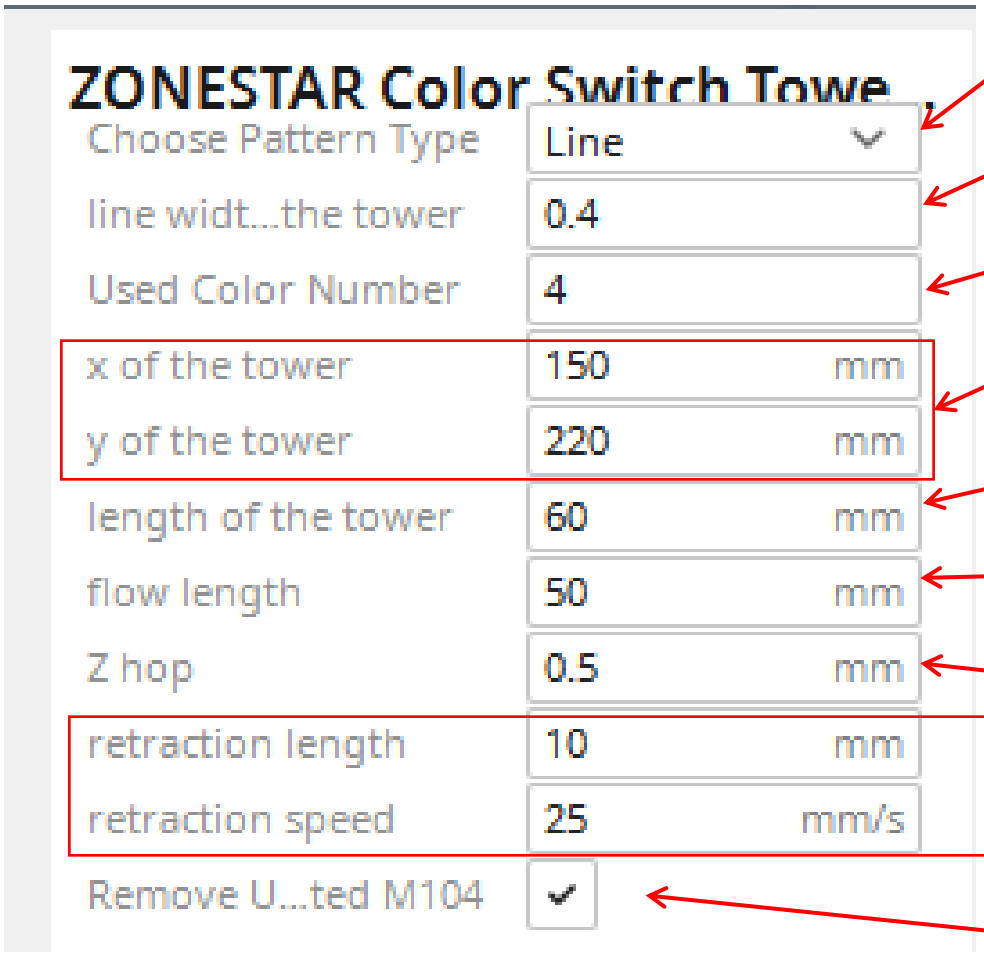




Slicing multi colors 3d object - Genarate color switching tower

2.Modify tower parameters.

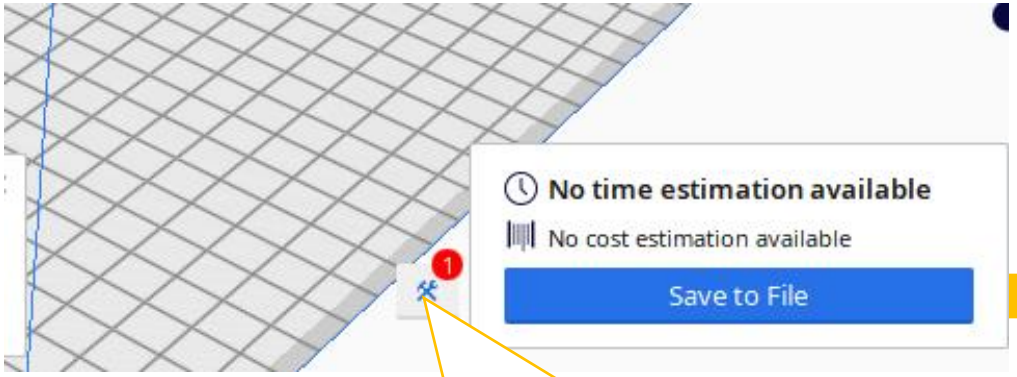
NOTE:The relevant parameters must be consistent according to the printing model and slicing software settings.



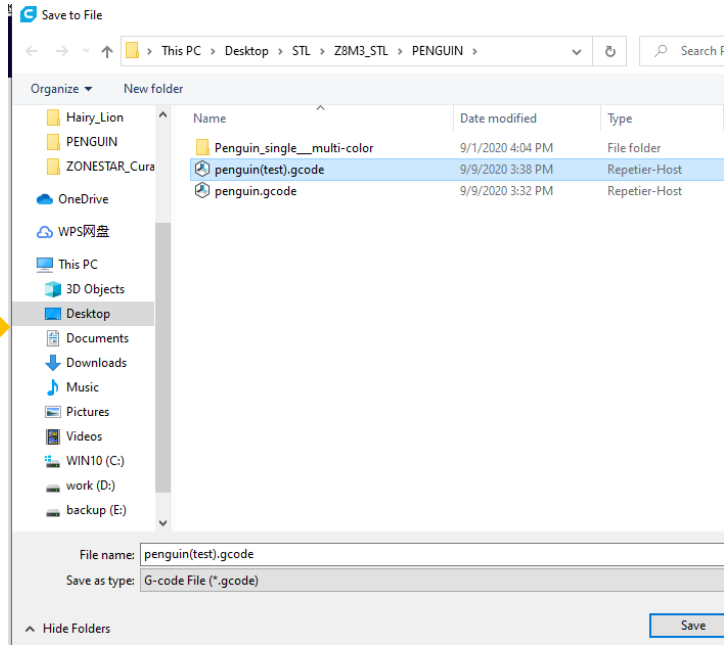
Pattern type of tower, you can choose line or square in current version
Pattern line width Recommended value is between 1 and 1.3 times of the nozzle size
How many colors (extruders) used in your gcode file.
the tower position (X and Y), choose the space of the printing platform and try to be close to the print
The length (X axis direction) of tower, recommend value is 60mm for line pattern and 40mm for square pattern
it will automatically exit the tower when feed filament over this length
Z axis hop how many distance when go to the tower and left the tower
Retraction length and speed, must set to the same with the settings in slicing(refer to page 13)
Cura will set the nozzle temperature when swith exturder, because mixing color hotend share one heater and temeprature sensor, It will cause issue on nozzle heating, so we need to remove it

Slicing multi colors 3d object - Genarate color switching tower

3.Open the sliced gcode file(e.g.: penguin.gcode) and then save it to a new gcode file (e.g.: penguin(test).gcode).



The plug-in tool is shown here.

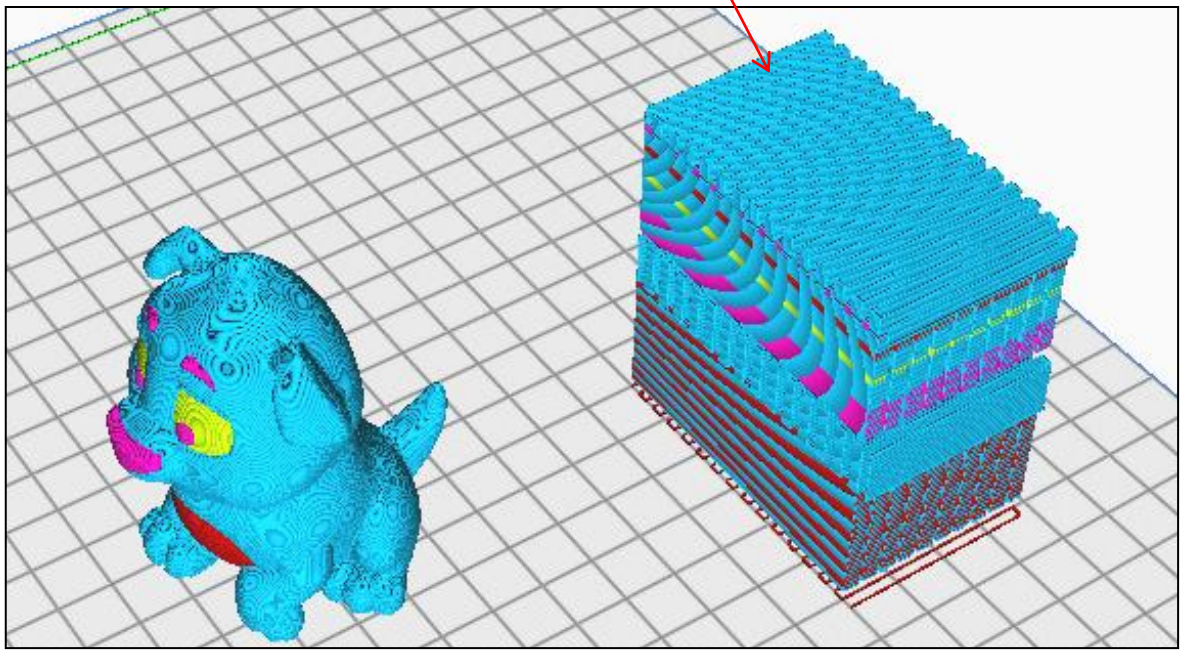


When save the new gcode file, the plug-in will work automatically and add the color swithing tower to the new gcode file.

Slicing multi colors 3d object - Genarate color switching tower

Open the new gcode file (penguin(test).gcode) you saved, preview it and check if the color switch tower has been generated and it is on the right position.

there is a color swithing tower has been added to the new gcode



Slicing more colors 3d object by using virual extruder

Slicing multi colors 3d object - Genarate color switching tower